



15

Road Cycling



Marc Quod

Performance Demands of Road Cycling

This chapter describes the sport of professional road cycling, discussing the various factors of importance for success, and how high-intensity interval training (HIIT) contributes to a cyclist's physical development.

Sport Description and Factors of Winning

One of the most universal sports in the world, road cycling is most popular in Western Europe, centered historically in France, Spain, Italy, and the low countries. Since the mid-1980s, the sport has diversified with professional races now held on all continents of the globe, all governed by the Union Cycliste Internationale (UCI). Road cycling incorporates a large range of competition types, with standalone, single-day races being competed over distances of 3 to 300 km, or a series of races competed over 2 to 21 consecutive days. As well as the UCI's annual World Championships for men and women, the biggest event is the Tour de France, a 3 wk race that can attract more than 500,000 roadside supporters a day. Success in road cycling is influenced by a range of factors including terrain, environmental conditions, tactics, and strategy. However, the primary

predictor of performance in a cycling road race is the cyclist's physical capacity. Similar to rowing, swimming, running, and triathlon, success is largely determined by an athlete's physiological capacity over skill or technique. In road cycling, the physiological traits of primary importance are those related to endurance capacity, such as the interrelated qualities of maximum aerobic power, economy, and fatigue resistance.

Due to such large variation in the duration and topography of races, there is equal variation in the type of physiology that is successful in the different events (figure 15.2). As such, many different athlete phenotypes can be competitive in road cycling, as different phenotypes are advantageous in different aspects of the race. Contrasting examples of this are a mesomorphic "sprinter" who relies on an ability to produce a very high peak power output for a very short period of time and an ectomorphic "climber" who needs to maintain a high power output for a much longer duration.

One aspect common to both of these examples, and the reason why power output is expressed in relative terms (W/kg), is the significant impact of mass/gravity on performance. For both the sprinter and the climber alike, their ability to overcome the effect of inertia/gravity is influenced by their body mass, making this a critical part of cycling performance.



Figure 15.1 The position of the cyclist on the three axes illustrates the relative importance of the three main physical capacities for elite participation, acknowledging this varies slightly between rider type. Outside of critical psychological aspects, which are difficult to quantify, the pie chart shows the general relative importance of skills (5%), tactical awareness (25%), and physical capacities (70%) for cycling success.

Adapted from G.A. Nader, “Concurrent Strength and Endurance Training: From Molecules to Man,” *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.



a



b

Figure 15.2 Contrasting examples of (a) Marcel Kittel, a mesomorphic sprinter; and (b) Chris Froome, an ectomorphic climber.

a. Chris Graythen/Getty Images; b. Justin Setterfield/Getty Images.

Road cycling is not without its skill requirements. Descending narrow mountain roads at speeds of up to 100 km/hr requires balance, cornering, and braking skill as well as focus. The ability to ride and navigate “road furniture,” dirt and cobbled sections of a course within a fast-moving peloton of up to 200 riders while simultaneously making decisions about the race situation/tactics, requires quick reflexes, fast thinking, and sustained concentration

over a long event. These aspects of the sport clearly require practice, skill, and experience, but compared with sports such as American football or tennis, the requirement for skill is comparatively low, making road cycling a sport primarily about the athlete’s physiology.

In addition to physiological capacity, there are other important factors that play a role in road cycling success. These include:

1. **An ability to quickly adapt and roll with the punches.** Cycling events take place throughout the year and in different parts of the world. Consequently, environmental factors such as altitude, snow, rain, heat, wind, and cold (sometimes each within a single race) are significant external challenges.
2. **There's a kingpin, but it's a team sport.** A cycle race is competed between teams but has an individual winner, and the winner typically depends on the support of the team to achieve success.
3. **Multiple teams on the pitch make things interesting.** Most team sports are adversarial (team A versus team B), however, in a professional cycle race, 15 to 25 teams of 5 to 9 riders compete at the same time.
4. **Prizes galore add further to complexity.** Within a given cycle race, there are multiple classifications to compete for. In addition to the race winner (first across the line), competitions for points (sprint, mountain, combination) as well as team classifications and total accumulated time (in multiday events) are included, making cycling a game of chess on wheels.

Each of these factors creates an array of incentives and temporary alliances that impact tactics and strategy. As a result, the specific physical requirements that are successful in any given event can vary greatly. The competition requirements for a sprinter in athletics or a rower are clearly established before the competition and do not change. This is not necessarily the case in professional road cycling. As such the specific goal of training depends on an individual rider's strengths and weaknesses, the type of race, and terrain being prepared for, and the likely tactic or strategy to be employed by the team. A benefit of HIIT training is that it can be infinitely varied to tailor the demand of training sessions to replicate the specific physiological requirements that are important in a cyclist's targeted events.

Physical Performance Contributions to Winning

Unlike many other physical capacity-based sports (athletics, rowing, swimming, and running), cycling is competed over much longer durations (up to 300 km for single-day events and ~3500 km across 21 consecutive days of racing in Grand Tours). It is an extreme endurance event (13); consequently, endurance capacity is a key performance indicator.

Figure 15.1 highlights these aspects for cycling in relation to the other sports discussed in this text. In addition, unlike other endurance-based sports, which require athletes to maintain a consistent, maximal-paced effort for a given duration, during a cycling race, the pace is variable. Road cycling is a dynamic event characterized by periods of sustained power output with alternating periods of high and low intensity, the combination of which can be infinitely variable (2).

Figure 15.3 highlights the variable competition demands encountered by a cyclist within and across different stages of a single race. During a flat/sprint stage, the power demands are particularly variable, requiring short maximal accelerations as well as accumulating substantial amounts of time at very low intensity (<100 W). However, during a mountain stage, the power output is less variable but incorporates long periods at sustained moderate intensities. This inherent variability means that every race that a cyclist starts has a unique set of physical demands.

Contrary to popular belief, it is not *necessarily* the cyclist with the highest $\dot{V}O_{2\max}$, or greatest sustainable power output, who wins a road race. The combined impact of tactics, drafting, constantly changing terrain, and associated variable energetic demands means the outcomes of mass start road races are typically determined at one or more critical periods of the races. These distinct periods require the cyclist to produce maximal power output for a race-specific duration. Therefore, depending on the nature of the race (figure 15.3), the winner can potentially be the cyclist who produces the lowest average power output over the duration of the race but produces a maximal effort that is greater than his or her competitors at a critical moment(s) (20). Such acute moments can vary greatly, being as short as 10 s in the final 100 m of the race, to a sustained 5 h effort in a solo breakaway, with every foreseeable combination in between (7, 16).

In summary, to be competitive, a road cyclist requires:

1. A large aerobic capacity, which enables prolonged medium intensity exercise, repeated over sequential days, and
2. An ability to produce a relatively high power output for race-specific periods, the duration of which is variable and may need to be repeated an undefined number of times.

Maximizing the necessary energy systems required to perform this large variation in competition

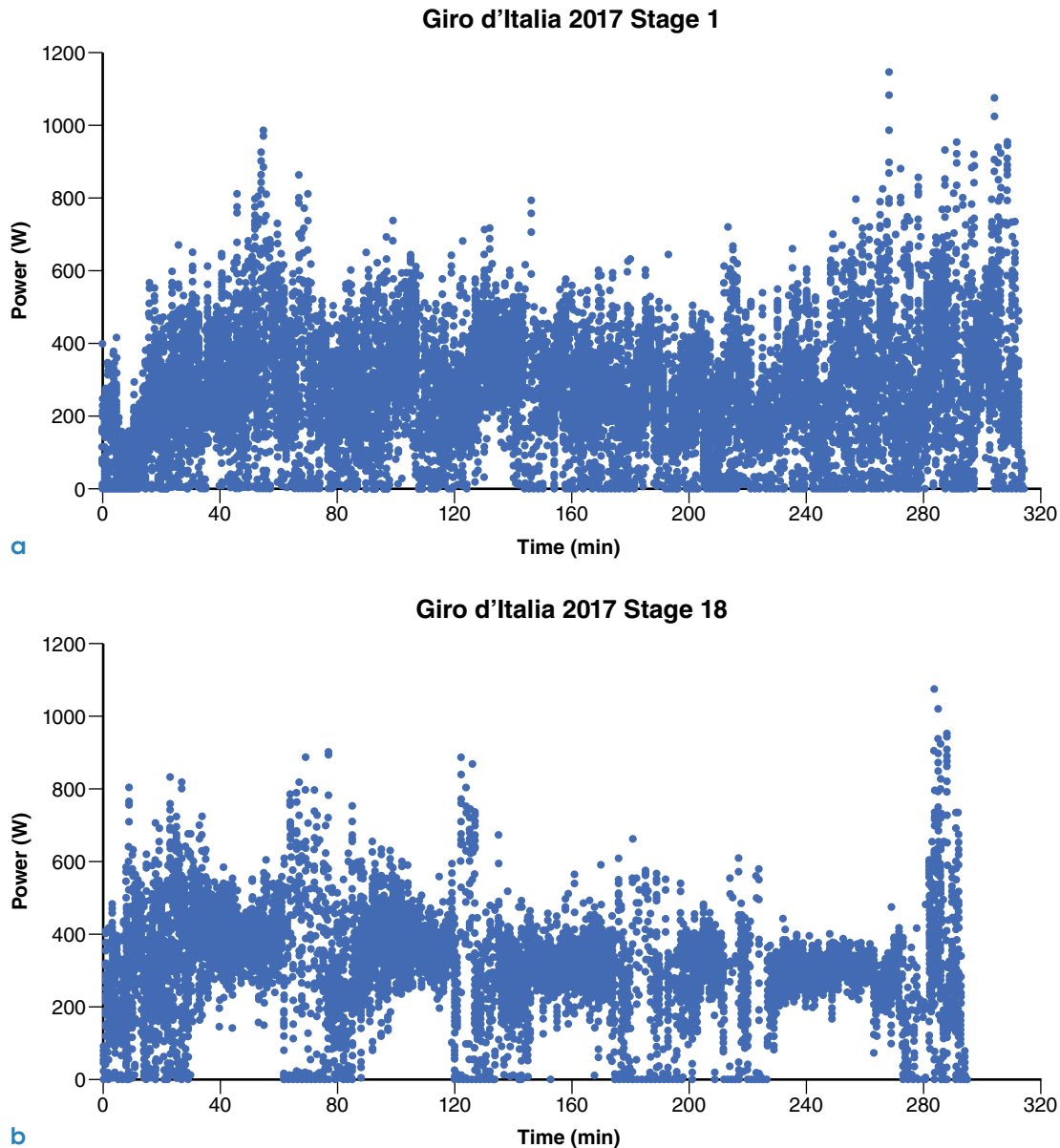


Figure 15.3 Power versus time for the same cyclist during two different stages in the 2017 Giro d'Italia: (a) a flat sprint stage (stage 1); (b) a mountain stage (stage 18). Data are 1 s averages, measured by an SRM power meter.

demands provides an interesting programming challenge for the cycling coach.

Targets of Physical Performance in Road Cycling

As discussed above, road cycling success is built primarily on an aerobic foundation, but anaerobic capacity often dictates the winner. Highlighting the

aerobic component of road racing, an analysis of more than 300 flat, semimountainous, and mountainous road races has shown that riders' heart rates average between 50% and 60% of HR_{max} , with average power output during flat and mountainous races reported as 220 W (3.1 W/kg) and 246 W (3.6 W/kg), respectively (18, 19). At first glance, these numbers appear relatively low. However, over the duration of a typical road race, a large number (>100) of very short (<15 s) efforts at power outputs greater than

70% of a rider's peak aerobic power are required (7). Combined with the typical long durations of such road races, these factors create very large energy expenditure and replenishment requirements (~25 MJ/d) (5). The mechanism enabling the expenditure of such large amounts of energy over consecutive days is the cyclist's endurance capacity, which is traditionally measured by maximal oxygen consumption ($\dot{V}O_{2\max}$) and/or maximal aerobic power (MAP). Cyclists tend to have very high values of $\dot{V}O_{2\max}$ and MAP with typical $\dot{V}O_{2\max}$ values ranging between 75 and 85 ml·kg⁻¹·min⁻¹ and MAPs of substantially greater than 6.5 W/kg. Chris Froome, the winner of the Tour de France in 2013, 2015, 2016, and 2017, has a reported $\dot{V}O_{2\text{peak}}$ of 5.9 L/min (84 ml·kg⁻¹·min⁻¹), which occurs at a power output of 525 W (7.5 W/kg) (4).

While a large aerobic capacity is a requirement for success, this feature alone is not what separates good cyclists from the best. Comparing submaximal thresholds (see chapters 3 and 4) of elite and professional cyclists, both aerobic (i.e., first ventilatory threshold; VT_1) and maximal lactate steady-state/critical power (i.e., or second ventilatory threshold; VT_2) thresholds occur at significantly higher exercise intensities, with submaximal lactate levels substantially lower in professional compared to elite-level cyclists (14). These physiological adaptations enable cyclists to repeatedly sustain relatively high percentages of their $\dot{V}O_{2\max}$ (>85%) for prolonged periods of time (30 to 60 min) within a road race. In addition, the lowered physiological disturbance at submaximal intensities enabled by high thresholds enhances fatigue resistance and a cyclist's ability to recover both within a race and across multiple days.

It has been highlighted that race-winning moves in road cycling are typically much shorter than the event itself. These moments can vary infinitely in duration, intensity, and pacing pattern, but can be grouped into three general categories:

1. **Sprints:** a maximal sprint in the final meters of a race, which can be as short as 10 s at >17 W/kg (17);
2. **Short attacks:** a repeated series of efforts, designed to distance rivals before reaching the finish line, of between 30 s and 5 min at 7 to 10 W/kg;
3. **Sustained efforts:** a maximally paced effort, typically over the final 10 to 60 min of a race at 5.5 to 7 W/kg (13).

Naturally, these effort types do not occur in isolation. Throughout a race, riders will be required to produce an unknown combination of each of these types of efforts. Cycling is clearly dominated by aerobic and anaerobic capacities, and as the efforts described above occur with relatively slow muscular contractions, the neuromuscular component of training is small compared to many other sports. While this aspect of training should not be overlooked for any athlete, the focus of HIIT training sessions for cyclists tends to be on developing aerobic and anaerobic metabolic capacity.

Cyclists in a professional team are not assigned positions as is the case in most team sports. Rather, their individual physiology (see figure 15.2) and relative ability to perform each of these three types of effort determines the role that is undertaken in each race. Corresponding with the three general categories of effort, there are three general types of rider:

1. The sprinter: Sprinters such as Marcel Kittel, Andre Greipel, Mario Cipollini, and Mark Cavendish target the final 200 m of a (normally flat) race. Up to this point they are looked after by their teammates with the goal to expend as little energy as possible throughout the race, saving this for a maximal acceleration in the final meters of the race. The sprint is typically between 10 to 15 s in duration and at power outputs of greater than 17 W/kg (17). Consequently, training tends to focus on short, high-intensity efforts that maximize the athlete's ability to deplete energy stores rapidly. However, while anaerobic adaptations are the focus for these riders, sprints occur at the end of 3 to 7 h of racing and across consecutive days. In addition, the ~10 min prior to the final sprint becomes progressively more difficult and requires a series of near maximal sprints to maintain position in the peloton (17). Therefore, to be fit enough to arrive at the finish line with the competition, aerobic and fatigue resistance aspects of fitness cannot be overlooked.

2. The all-rounder: As the name implies, these types of riders may not have a single aspect of performance that they are the best at, but rather, are very capable in many different situations. Riders such as Peter Sagan, Philippe Gilbert, Paolo Bettini, and Eddy Merckx are famous examples of an all-rounder. As these riders are capable across varied terrain, the focus of training will depend on their goal in upcoming races and will span the breadth of aerobic and anaerobic training variables. Due to the versatile strength of these types of riders, they

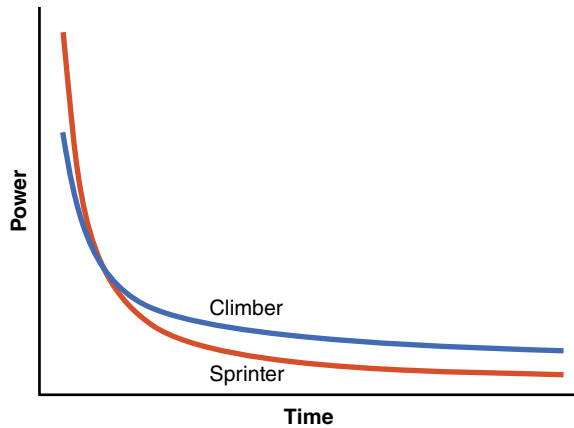


Figure 15.4 A generic power profile that illustrates the relative differences in ability to produce power for two different types of cyclist (climber versus sprinter). For short durations of time, the sprinter can produce very high power outputs, however, as the duration of the effort extends, the relative power produced by the climber exceeds that of the sprinter. The power profile of the all-rounder will often fall between these two extremes.

are often the ones called upon to work during a road race. This can include riding at the front of the peloton for multiple hours at medium intensity to maintain a specific time gap to the breakaway group, returning to a following support vehicle to collect food and drinks for the team leaders, or riding in front of a team leader throughout a race, providing him with an aerodynamic draft and easy ride in the peloton. In these situations, the all-rounder's job is to expend his energy so that the team leader can conserve his own.

3. The climber/general classification (GC): The climber is at the opposite end of the spectrum to the sprinter (as illustrated in figures 15.2 and 15.4). Climbers such as Chris Froome, Alberto Contador, or Nairo Quintana lack the ability to produce very high absolute power outputs required to win a bunch sprint. Rather, they excel by repeatedly sustaining high relative workloads as they ascend mountains. Races typically classified as mountainous include routes that can accumulate up to 5500 m of vertical ascent requiring the climber to accumulate up to 2.5 h at intensities greater than 75% of $\dot{V}O_{2\max}$. Consequently, maximizing aerobic capacity and the sustainable percentage of that capacity are the primary targets of training sessions for the climber.

Key Weapons, Manipulations, and Surveillance Tools

For a professional cyclist, the majority of training hours (10 to 30 h per week) are spent accumulating volume at low intensities (50% to 80% of $HR_{\max}$, 40% to 70% $\dot{V}O_{2\max}$). This relatively unstructured part of the training program is important as it replicates the volume of exercise required in races, as well as positively influencing body composition, economy, and aerobic capacity (10, 12). In addition to the large amount of time spent at low intensities, cyclists also complete a substantial amount of time training at medium, or mid-zone, intensity. Typical aerobic/medium intensity intervals are completed over durations of 15 to 60 min at intensities of 80% to 90% $HR_{\max}$ or 70% to 85% $\dot{V}O_{2\max}$. Given the event requirements, this is the aspect of fitness and training that cyclists prioritize.

For most cyclists, HIIT contributes considerably less time to the training schedule (<10%) compared to low- and medium-intensity training. However, even though it contributes only a small percentage of total training time, HIIT remains a critical component of the training program. Road races are won (or lost) in a moment of maximum effort, and HIIT is particularly important for maximizing the adaptations of both oxidative and nonoxidative energy systems that fuel these maximal efforts (see chapter 3). The structure of the various HIIT intervals that are relevant to cycling performance varies greatly, along with the diverse range of competition requirements within road races. The key weapons, or HIIT formats, including their manipulations and surveillance (monitoring) tools, are described next.

HIIT Weapons, Targets, and Manipulations

As shown in figure 15.5, the most common HIIT formats or weapons included in cycling training programs primarily include long interval, short interval, and sprint interval training, aimed at type 1 through 5 targets, with a metabolic emphasis. Repeated sprint training, which incorporates very short duration efforts (3 to 7 s), are used infrequently by road cyclists.

Long Intervals

Long HIIT intervals usually take the form of 1 to 6 min work durations at intensities of 85% to 95%

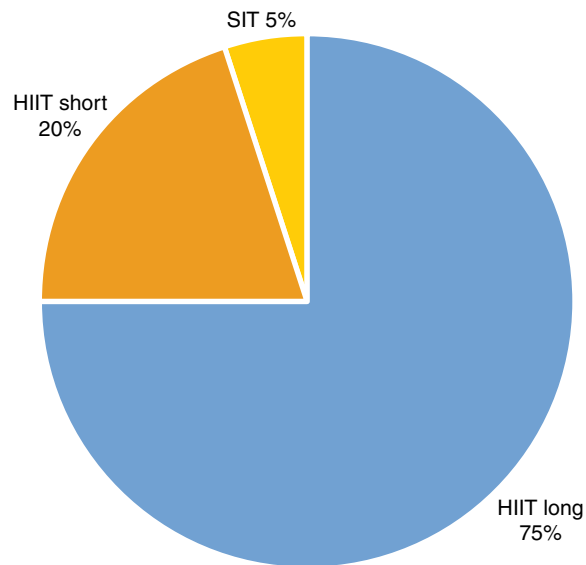


Figure 15.5 Percentage of the different HIIT formats (weapons) used throughout the annual season in professional cycling.

of HR_{max} , 85% to 100% of $\dot{V}O_{2max}/MAP$, generally with work:rest ratios between 1:0.5 and 1:2. Total accumulated duration at these intensities during such interval sets is typically 20 to 40 min split between 1 to 5 sets. A traditional and common long HIIT interval session for a cyclist is 5 × 5 min intervals at 80% to 90% MAP with recovery intervals of 5 min at a very low intensity. The desired metabolic load for these HIIT sessions includes type 2, 3, and 4 targets. These different targets can be achieved by manipulating the work:rest ratio. For example, reducing the work period (<90 s) and increasing the rest period (>90 s) will focus the session toward using more anaerobic metabolism. Conversely, increasing the work period (>3 min) while reducing the recovery period (<3 min) will emphasize use of aerobic metabolism.

Short Intervals

Short HIIT intervals for road cyclists usually take the form of 20 to 60 s work durations at intensities of 100% to 140% of MAP, generally with work:rest ratios between 1:0.5 and 1:1. Total accumulated work duration for these interval sets is generally between 10 to 20 min, split between 3 to 10 sets. An example of a short HIIT session for a cyclist would be a series of 6 × 6 min intervals of alternating 20 s at 110% MAP with 10 s at very low intensity, with recovery between the series of 4 min. The desired metabolic load for these HIIT sessions includes type 1, 2, 3,

and 4 targets. Similar to long HIIT intervals, these different targets can be achieved by manipulating the work:rest ratio. Alternatively, the intensity of both the work or rest period can be adjusted to shift the balance between anaerobic and aerobic metabolism (see chapter 5).

Sprint Interval Training

Sprint intervals for road cyclists tend to be the domain of the “pure” sprinters, although they are relevant to all cyclists who want to improve their sprints. Sprint intervals are composed of maximal efforts between 10 and 30 s, with recovery periods of 2 to 10 min. As the goal of these intervals is to reach maximal values, the number of sprints is typically low (1 to 5), and time between sets relatively long (>30 min). The desired load for these HIIT sessions includes type 5 targets.

Manipulations of Interval Training Variables

A marathon runner is acutely aware of the precise physiological requirements of his next race and HIIT interval sessions can be tailored and periodized appropriately for the specific energy system requirements of a 2 h marathon (see chapter 14). However, the competition demands of a road race vary considerably from race to race (as highlighted in figure 15.3) and are not always predictable. Consequently, each of the 12 different HIIT variables (chapter 4) are important factors in each HIIT session and can be manipulated depending on the type of race a cyclist is targeting.

Physiological parameters such as critical power (CP), onset of blood lactate accumulation (OBLA), MAP, and lactate and ventilatory thresholds are often utilized to guide HIIT interval sessions. In cycling, the use of power meters built into the bicycle, combined with measurement of heart rate, has allowed for very precise prescription and measurement of exercise intensities relative to various physiological markers (see chapter 8). With this technology, the cyclist is able to ride at a specific power output that is associated with the physiological parameter of interest. The advantage of this approach to HIIT interval prescription is that it allows the coach to focus on and isolate the underlying physiology and energy systems that support a given competition demand.

In addition, similar to the track-and-field approach to the prescription of HIIT sessions (chapter 2), the

expected competition demands of an upcoming race can also be used to develop specific HIIT training sessions. Using GPS technology and historical records of the speed/power relationship over different sections of terrain, it is possible to make predictions as to the critical sections of upcoming races. For example, a circuit race that includes 15 repetitions of a 1 km climb will require repeating ~2.5 min efforts at >6 W/kg each lap. This critical section of the race can be used to design an HIIT training session. While the components of these sessions are manipulated without reference to the body's energy-delivery systems, the advantage of this method is that the HIIT sessions replicate the specific competition requirements.

The primary factors of any HIIT session include the intensity and duration of the work and rest periods. Typically, as most HIIT work intervals are relatively short in duration, power (W) is best used to prescribe the intensity. There are no required or recommended durations that should be incorporated in a cyclist's HIIT training session, rather the choice of appropriate duration should be anchored by either upcoming competition demands or the limits of the relevant energetic systems. Similarly, the number of intervals in a series, the series duration and time between each series of HIIT intervals will depend on the specific competition demand or the physiologic or metabolic system being targeted by the coach.

Modality is also an important HIIT variable in cycling. Typically, this component of an HIIT session is manipulated by altering terrain (positive, negative, or neutral gradient) and cadence (rpm).

Choice of terrain for HIIT sessions is primarily dictated by the type of terrain that the rider will encounter in upcoming races. However, a secondary benefit of utilizing different gradients in training is the ability to impact the inertial load encountered by the rider. As shown in figure 15.6, due to the disparity in speed while climbing versus riding on the flat, for a given power output, the inertial load during these activities and the associated self-selected cadences are different (9).

Similarly, manipulating the cadence (rpm) during a work interval will impact the balance between muscular strength and coordination as well as the neuromuscular and metabolic demands. For example, a 5 min work interval completed at 80% of MAP and 60 rpm will differ from an interval done at the same intensity but with 120 rpm. This is because power is a product of torque and angular velocity. While the power remains constant across these two

work intervals, due to the change in cadence (angular velocity), the torque requirements in the low-cadence condition (60 rpm) are double that of the high-cadence condition (120 rpm). Using cadence to manipulate the torque requirements of a work interval is analogous to focusing on strength versus speed in the gym (see chapter 6).

Because road races are not constantly paced efforts, there is also opportunity to add additional layers (complexity) to HIIT interval sessions to replicate race demands. Specifically, levels of intensity during the work and rest periods can be varied. An example may include a period of moderate-hard intensity work immediately prior to a maximal sprint in a sprint interval training session. This replicates the demands of a sprinter in a race situation, fighting to maintain good position in the peloton in the lead up to her final maximal acceleration. Alternatively, periods of medium-hard intensity may follow a maximal effort prior to the rest period. This type of effort replicates the demand of an all-rounder who needs to accelerate maximally to distance himself from the peloton midway through a race, but then needs to maintain a moderate intensity to maintain this distance to the finish line.

Finally, as training sessions tend to be relatively long in duration (2 to 7 h), the decision of where to place an HIIT interval within a training session is important. Placing HIIT intervals in the beginning, middle, or end of a session changes the initial state of fatigue and energy availability (glycogen stores) that the interval session is commenced with (chapter 4). An athlete who is looking to increase absolute power production in a sprint may choose to complete sprint-specific HIIT intervals in the first hour of a training session to maximize quality and achieve peak power outputs, whereas a rider looking to improve fatigue resistance may complete the intervals in the final hour of training when fatigue is relatively high and glycogen stores are already depleted.

Surveillance and Monitoring Tools

The primary surveillance and monitoring tools used by professional cyclists are the power meter, heart rate monitor, and their associated analysis software. Other useful monitoring tools include a cycling computer that records speed, time, and distance as well as the vertical distance accumulated. Databasing and analyzing trends in these variables over time allows an athlete to precisely track progress,

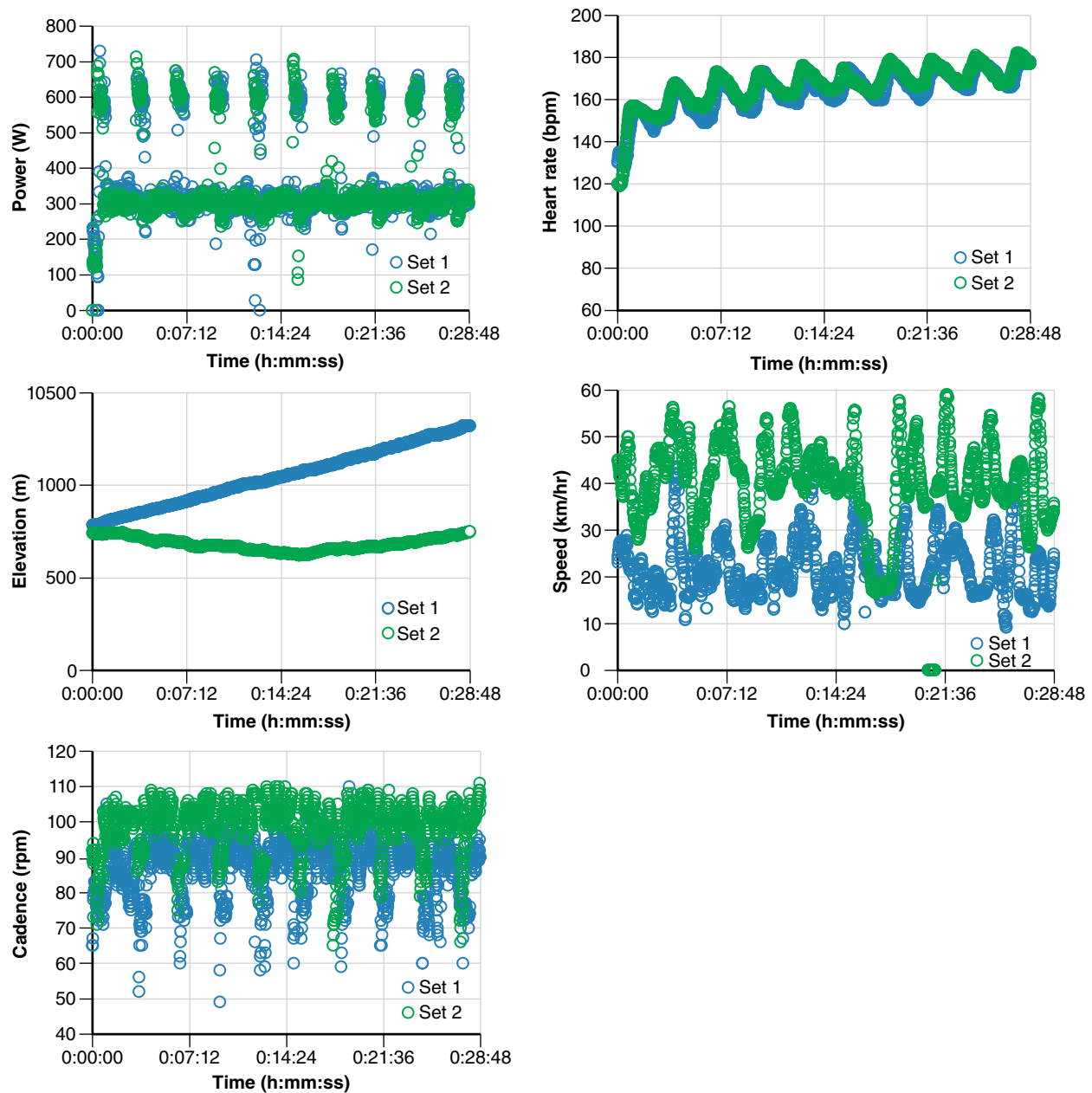


Figure 15.6 Two HIIT interval sets, both 10×30 s at 120% MAP, 2.5 min at 60% MAP. However, set 1 is completed on a climb (average gradient of 5%), whereas set 2 is completed on level ground. Although the power output and heart rate response are almost identical between the two interval sets, the influence of road gradient on speed and subsequently self-selected cadence, torque, and in turn, neuromuscular load, is different.

highlight breakthroughs in performance or aspects of training that have been overlooked.

Load

Intensity prescription based on heart rate is the most commonly used strategy to control and monitor exercise intensity, though it is best suited to prolonged submaximal exercise. The high intensity and

relatively short duration HIIT intervals and the associated lag in HR response (see chapters 2, 8, and 9) preclude the use of HR for prescribing HIIT intervals. The best method for prescribing and monitoring intensity during HIIT sessions is with a power meter attached to the bicycle. These technological innovations allow for accurate and reliable measures of both power output (external work/stress) and heart

rate (internal work/strain), which makes cycling a privileged sport in the realm of exercise science and has allowed for numerous laboratory- versus field-based comparisons of performance (1, 8).

Training Status

With the use of power meters, during the course of training or racing, strengths and weaknesses of riders can be identified on any portion of the power-duration continuum where a maximal effort occurs (chapter 4; i.e., W' , CP). For example, markers of sustainable threshold power output are commonly used for road cyclists, and include a 20 or 60 min power output, popularly referred to as functional threshold power (FTP) (3).

The information derived from power meters, heart rate monitors, and their array of associated analysis technologies has proven useful in a number of applications for the cycling scientist and enthusiast. The ability to review a session after training by coach, support team, and athlete allows informed decisions with respect to training and recovery. When the data and analysis (such as the power:heart rate ratio) is compared alongside an athlete's rating of perceived exertion or informal comments, the combined information provides for a sound global indicator of an athlete's state of readiness to perform. Additionally, as much information is freely available today, many athletes research and learn this information and compare their power output numbers to published records. As a result, many athletes in the sport are informed and well versed when it comes to commonly used cycling-specific training lingo, including normalized power, power:heart rate ratio, training stress score, training stress balance (TSB), and chronic training load (CTL) (3).

Given the large training volumes, and the general belief that more training is always better, the risk of developing chronic overreaching is very real (see chapter 7). To help predict or minimize this risk, it is possible to accurately track long-term training load, however the predictive ability of various training load calculations (such as TSB) remains limited and tied to rules of thumb. Advances in other monitoring tools such as heart rate variability are helping (chapter 9), but again, their predictive power is limited.

Despite all this data and technology at our fingertips, there are still many unanswered questions in the world of road cycling. The systematic collection of reliable power and heart rate data has existed

for many years, yet we are still some way from understanding the optimal balance of training variables. The varying and diverse rules and goals of the sport, the inherent uncertainty of race demands, and the infinite uniqueness of individual athletes maintains this challenge.

Strategies for Structuring the Training Program Using HIIT

Implementing an HIIT program in the real world of road cycling is a complex task. As with most sports, there are numerous uncontrollable factors that must be considered.

Controllable and Uncontrollable Factors

The World Tour is the top tier of professional cycling and includes 37 races (175 individual race days) spread throughout the world including Australia, China, North America, the Middle East, and Europe. Professional teams consist of 24 to 30 riders and compete in the World Tour as well as other lower-level races each season. Individual riders typically complete 40 to 90 race days each year, spread primarily between January and October. Given the large number of races available, the competition schedule for an individual rider can vary from year to year. As cyclists compete in substantially more races than other endurance athletes, this variable and relatively large race program is the primary factor that impacts an individual's training program.

Due to the extreme nature of road races, the physiological load of a race is typically much higher than can be achieved in training. Therefore, appropriate periods need to be built in to training programs both before (taper) and after (recovery) each race to avoid overreaching and negative training outcomes. Depending on the context, these recovery periods can be 2 d to 2 weeks.

Finally, travel requirements are another challenge that road cycling teams must consider. Races are conducted all over the world, and training camps (typically in hot climates or at altitude; chapter 4) are often in difficult to reach places. Thus, recovery from travel stress as well as time for circadian adaptation to the time zone needs to be considered; otherwise, the health of the athlete may be compromised (chapter 7).

Program Structure and Progression

There are a number of different approaches utilized in cycling, and all have been used successfully at one time or another, from classic to block periodization models, and everything in between. As the competition program is irregular, and to some degree chosen by the athlete, there is vast scope for variation and individualization of the training program and its periodization. Due to high volumes of training in nearly all programs, appropriate progression of training load remains a key to avoiding overuse injuries in road cyclists. Despite this variation, some general observations can be made across the macro-, meso-, and microcycles observed in the majority of the road cycling peloton.

Macrocycle

A road cyclist's season is typically periodized over a 12 month cycle with key racing objectives or peaks lasting 3 to 6 wk. The best cyclists tend to have just one or two primary objectives each season. For these athletes, a classic Matveyev approach (high to low volume, low to high intensity) to periodization is common (15). The duration between season peaks allows for progressive training blocks with targeted physiological overloads. However, many in the peloton are required to be at a high level of fitness for sustained periods, often a series of three to five objectives separated by brief periods of 1 to 3 wk. For these cyclists, periodization can be similar to many team sport athletes who build a high level of fitness in the preseason and then maintain that through a long competitive period. The German athletics coach Peter Tschene has described this approach to periodization, where all aspects of fitness are trained at a high level at the same time by systematically varying each component of fitness (21, 22). Incorporating the use of frequent, short recovery periods enables the athlete to maintain a readiness to compete throughout a long competitive season.

Mesocycle

A typical basic or preparatory mesocycle in cycling follows the traditional 4 wk approach, with 3 loading weeks and 1 recovery week. However, this will vary depending on the available time between races and can include shock mesocycles (~1 or 2 wk),

precompetition or transition mesocycles (~2 to 4 wk), competition mesocycles (~3 to 6 wk), and recovery mesocycles (~1 to 4 wk).

Microcycle

The microcycles of a road cyclist's program tend to be variable. Whereas 2 and 3 day training blocks separated by a single recovery day are the most common and traditional, single day and up to 10 day microcycles are not uncommon. This variation is in line with the competition requirements of the sport. Cyclists who are targeting single-day races are more likely to include single-day training blocks with a focus on maximal expenditure of energy in a single day, whereas a cyclist targeting the overall classification in a 3 wk Grand Tour will incorporate longer microcycles to reflect the recovery and repetition aspects of competition demands.

Periodized Nutrition

As outlined in chapter 4, periodized nutrition also plays an important role in the life of a road cyclist. Not only is nutrition important in controlling a cyclist's weight but manipulating the fuel available for a training session can alter the adaptations to that session (6).

Carbohydrate is the most common macronutrient to be targeted as glycogen stores and availability can be easily influenced by the amount of carbohydrate consumed before, during, or after a training session (11). Performing training sessions with low glycogen stores or minimal exogenous carbohydrate availability (popularly termed *train low*) and consuming carbohydrate at a rate >100 g/h during a training session (popularly termed *train high*) are typical methods of carbohydrate manipulation (11).

Today, carbohydrate quantity is a programmed component of nearly every training session a rider does, and can be incorporated into HIIT sessions, low- and medium-intensity training sessions, as well as being a component of the overall periodization plan.

Sample Training Programs

An example of a cyclist's training program, including planning during the preparation and competition phases alongside HIIT integration and target types, are shown in figure 15.7.

Macro-cycle	10	11	12	13	14	15	16	17	18	19	20
Phase	Preparation 2					Transition 1					Competition 1
Race		Training camp	Training camp			Altitude camp	Altitude camp	Altitude camp			Tirreno-Adriatico
Meso-cycle	1	2	3	4	R	1	2	3	R	1	2
Mon	Medium intervals		Type 1		Volume	Acclimation	Volume	Aerobic intervals			
Tues	Strength	Aerobic intervals	Strength	Type 1		Acclimation					
Wed	Volume	Medium intervals	Volume	Medium intervals		Volume	Strength	Type 2		Medium intervals	
Thu	Aerobic intervals	Strength	Aerobic intervals	Strength		Volume	Medium intervals	Volume			
Fri		Volume		Volume			Volume			Recovery	
Sat	Type 1		Medium intervals			Strength		Type 2	1 day race	Recovery	
Sun	Volume		Type 1	Aerobic intervals		Aerobic intervals	Type 2		1 day race	1 day race	

Macro-cycle	21	22	23	24	25	26	27	28	29	30	31
Phase	Competition 1 (continued)		Transition 2						Competition 2		
Race	Tirreno-Adriatico	Volta Catalunya	Altitude camp	Altitude camp	Altitude camp	Altitude camp	Altitude camp		Giro d'Italia	Giro d'Italia	Giro d'Italia
Meso-cycle	3	4	R	1	2	3	R	R	1	2	3
Mon				Strength + volume		Medium intervals					
Tues				Volume							
Wed				Medium intervals	Type 3 + strength	Type 2	Medium intervals				
Thu					Medium intervals	Medium intervals					
Fri				Strength + volume							
Sat	Medium intervals			Volume	Type 2	Type 3 + strength	Aerobic intervals				
Sun				Medium intervals	Volume	Volume	Volume				

Figure 15.7 Preparation phase of a cyclist's training plan. Details of individual HIIT sessions are: long-interval HIIT type 1 = 5 x 5 min at 85% MAP (5 min easy); short-interval HIIT type 2 = 3 sets of 10 to 16 x (35 sec at 110% MAP, 100+ rpm [25 s at 60% MAP]); long-interval HIIT type 3 = 2 sets of 6 x 3 min at 90+% MAP (3 min easy).

CASE STUDY

The Team Time Trial World Championship is an event competed over distances of 30 to 60 km (45 to 70 min depending on the terrain) in teams of six riders. Each rider takes a turn at the front of the group while the other riders follow in the lead rider's draft. The goal of the event is to get at least four of the team of six riders from point A to point B in the least amount of time.

Historically, to win this event requires an average speed of 52 to 57 km/h. Depending on the composition of the team and the terrain, riders take turns at the front of the group for 20 to 90 seconds. The power requirement while riding at the front of the group at these speeds is roughly 110% to 120% MAP. After each rider's turn at the front, the rider moves to the back of the line of riders, rotating through to the front sequentially. While following teammates, the power output in the aerodynamic slipstream is reduced significantly. The second rider in line is required to produce ~80% of the power produced by the lead rider, third ~65%, and fourth to sixth ~55%, typically around 60% to 70% MAP. Although dictated primarily by the terrain, the length of time between turns at the front is typically 2 to 5 min. As you can see, the Team Time Trial is essentially an HIIT session, alternating high-intensity work periods with lower-intensity recovery periods.

In an ideal situation, preparation for the Team Time Trial involves the selected six riders training together over a course that replicates the race. However, due to other team objectives and the proximity of other races, it is not always possible to get all six riders together prior to the event. In 2016, our team had riders coming from very different backgrounds: two riders had recently competed in the Olympic Games in the Team Pursuit (an event that lasts <4 min), three had completed the Tour de France (>3300 km across 3 wk, 8 wk prior to the event) and one had completed the Vuelta España (>3100 km across 3 wk, 3 wk prior to the event). The diversity in training and racing in the lead-up to the event meant that each of the six riders had different recovery and training requirements. In addition, different race programs immediately prior to the event precluded any group training sessions.

The two riders coming from the Olympic Games and Team Pursuit training had already completed a lot of work to maximize their ability to produce power over short durations (<4 min), but the 2016 Team Time Trial World Championships were expected to take ~45 min to complete the 40 km course. Conversely, the remaining riders had recently completed Grand Tours and had spent 80+ hr over 3 wk at a medium intensity; they were undoubtedly fit but lacked the exposure to the specific high-intensity work required for the Team Time Trial.

As the riders coming from the Olympic Games lacked the required aerobic adaptations necessary to repeat efforts over a 45 min event, type 1 HIIT sessions, short intervals of 10 to 30 s at 110 to 130% MAP, were targeted in their training. Conversely, the riders coming from the Grand Tours needed to improve their ability to produce a high-power output while at the front of the team. Consequently, type 3 HIIT sessions, long intervals of 60 to 90 s at 90% to 110% MAP, were incorporated in their training.

Unfortunately, the team was not able to win the World Championships in 2016, but with the help of individualized HIIT training sessions, a group with limited opportunity to train together and from vastly different training backgrounds achieved a podium result, finishing within 1.5% of the winner's time.

This experience highlights the value in the inherent variability of HIIT sessions, which provided a distinct advantage in this situation. While the optimal way to prepare for such an event may be to have the whole team training together, by targeting and manipulating each of the 12 HIIT variables (see figure 4.2), it is possible to bring a diverse group of athletes together and be competitive as a team.

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